

G.7 continued...

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$$A = \int \frac{dx}{\sqrt{1-4x^2}}$$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C$$

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1} \frac{u}{a} + C$$

$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right| + C$$

$$u = 2x \Rightarrow du = 2 dx$$

$$\Rightarrow dx = \frac{du}{2}$$

$$\Rightarrow A = \int \frac{\frac{du}{2}}{\sqrt{1-u^2}} = \frac{1}{2} \int \frac{du}{\sqrt{1-u^2}}$$

$$= \frac{1}{2} \sin^{-1} u + C$$

$$= \frac{1}{2} \sin^{-1} 2x + C$$

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$$A = \int \frac{\sec^2 x \cdot dx}{\sqrt{1-\tan^2 x}}$$

$$u = \tan x \Rightarrow du = (\tan x)' \cdot dx$$

$$\Rightarrow du = \sec^2 x \, dx$$

$$A = \int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C$$

$$= \sin^{-1}(\tan x) + C$$

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$$A = \int \frac{\sin \theta}{\cos^2 \theta + 1} d\theta$$

$$u = \cos \theta \Rightarrow du = -\sin \theta \cdot d\theta$$

$$\Rightarrow -du = \sin \theta \cdot d\theta$$

$$A = \int \frac{-du}{u^2 + 1} = - \int \frac{du}{u^2 + 1}$$

$$= -\tan^{-1} u + C$$

$$= -\tan^{-1}(\cos \theta) + C$$

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$$A = \int_1^{\sqrt{3}} \frac{\sqrt{\tan^{-1} x}}{1+x^2} \cdot dx$$

note:

$$(\tan^{-1} x)' = \frac{1}{1+x^2}$$

$$u = \tan^{-1} x \Rightarrow du = (\tan^{-1} x)' \cdot dx$$

$$= \frac{1}{1+x^2} \cdot dx$$

x	1	$\sqrt{3}$
$u = \tan^{-1} x$	$\pi/4$	$\pi/3$

$$A = \int_{\pi/4}^{\pi/3} \sqrt{u} \cdot \frac{1}{1+x^2} \cdot dx$$

$$= \int_{\pi/4}^{\pi/3} \sqrt{u} \cdot du$$

$$= \int_{\pi/4}^{\pi/3} u^{1/2} du = \frac{u^{1/2+1}}{1/2+1} \Big|_{\pi/4}^{\pi/3}$$

$$\approx .25 \dots$$

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$$A = \int_1^3 \frac{dx}{\sqrt{x}(x+1)}$$

$$u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}} \cdot dx = \frac{1}{2} \cdot \frac{dx}{\sqrt{x}}$$

$$\Rightarrow u^2 = x$$

x	1	3
u	1	$\sqrt{3}$

$$A = \int_1^{\sqrt{3}} \frac{1}{x+1} \cdot \frac{dx}{\sqrt{x}}$$

$$= \int_1^{\sqrt{3}} \frac{1}{u^2+1} \cdot 2du = 2 \int_1^{\sqrt{3}} \frac{1}{u^2+1} du$$

$$= 2 \cdot \tan^{-1} u \Big|_1^{\sqrt{3}}$$

$$= 2 \left(\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right)$$

$$= 2 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = \frac{\pi}{6}$$